

FlyByWire feature user manuals

One manual per `core/*` branch of the fork (28 branches): what each feature does, how to use it, how to test it in the sim, what to expect, and its limitations with the reason

Aircraft	A32NX and A380X
Where to find it	This folder: each manual as HTML (with its <code>img/NN/</code> pictures) and as PDF; <code>FBW-user-manuals.pdf</code> holds them all, <code>FBW-user-manuals.docx</code> is the editable Word version
Branch	The <code>core/*</code> branches of the fork <code>TCdr/aircraft</code> : what to open the upstream pull requests from
Status	Restructured on 2026-10-04 from the 31 per-PR manuals of 2026-09-28 (their content is merged into the core manuals); each feature carries its own test status.
Pictures	Instrument pictures come from the local browser harness running the installed instrument bundles; the native WASM displays (weather radar, terrain, VD) are in-sim screenshots; audio features use diagrams. Features added after 2026-09-28 are described in text.

1. The manuals

2. Stacks: what must be merged first
3. Companion changes in aircraft-large-files

4. How to read them

1. The manuals

#	Manual	Aircraft	Branch	Based on	Features
1	Weather radar and terrain on the ND and VD (PDF)	A32NX and A380X	<code>core/nd-weather-terrain</code>	<code>master</code>	8
2	Surveillance: SURV controls, ECAM SURV lines and TCAS (PDF)	A380X (SURV panel, VD messages, ECAM lines) and A32NX + A380X (TCAS)	<code>core/surveillance</code>	<code>master</code>	7
3	Independent CPT and F/O MCDUs (PDF)	A32NX	<code>core/a32nx-mcdu</code>	<code>master</code>	3
4	ECAM procedures, EMER CANC and FIRE TEST (PDF)	A380X (ECAM, FWS) and A32NX + A380X (FIRE TEST option)	<code>core/ecam-fws</code>	<code>master</code>	5
5	Runway awareness: RAAS and OANS (PDF)	A32NX (RAAS) and A380X (OANS)	<code>core/oans-btv-raas</code>	<code>master</code>	4
6	A380X standby instruments (ISIS) (PDF)	A380X	<code>core/isis</code>	<code>master</code>	4
7	Lights: strobe and beacon timing, interior lighting, presets (PDF)	A32NX and A380X	<code>core/lights</code>	<code>master</code>	2
8	Fire warnings and the ENG FIRE pushbutton (PDF)	A32NX (fire warnings, LP fuel valve) and A380X (LP fuel valve)	<code>core/fire</code>	<code>master</code>	3

9	A380X MFD FMS and SURV pages (PDF)	A380X (the shared flight plan code of DIRECT TO, the 50 waypoint limit and the company winds is also in the A32NX FMS, unchanged for it)	core/a380x-mfd-fms	core/surveillance	16
10	A380X ATC COM and datalink (PDF)	A380X (the ACARS provider connection is shared with the A32NX)	core/a380x-atccom-datalink	core/a380x-mfd-fms	5
11	A380X fuel jettison and FINAL fuel (PDF)	A380X	core/fuel	core/a380x-mfd-fms	4
12	GPS receivers, GPS deselection and ECAM STATUS (PDF)	A32NX and A380X (shared systems code); the ECAM STATUS page and STS reminder are A32NX only	core/gps	core/a380x-mfd-fms	6
13	A380X interactive ND (PDF)	A380X	core/a380x-interactive-nd	core/a380x-mfd-fms	1
14	flyPad performance calculators (PDF)	A380X and A32NX (one shared flyPad screen per calculator; the company datalink reply time is A380X only)	core/efb-performance	core/a380x-mfd-fms	10
15	Cockpit printers (PDF)	A380X (printing, buttons) and A32NX (paper door, no torn paper edge)	core/printer	core/a380x-atccom-datalink	5
16	flyPad ground: taxi route, ATC frequencies and GSX (PDF)	A32NX and A380X (the OANS flags are A380X only)	core/efb-ground	core/efb-performance	5
17	flyPad Material 3 redesign and independent flyPads (PDF)	A32NX and A380X (the Buffet tab: A32NX only)	core/efb-redesign	core/efb-ground	15
18	A32NX air conditioning per the FCOM (PDF)	A32NX	core/a32nx-air-conditioning	core/gps	4
19	A380X cockpit buttons, lights and probes (PDF)	A380X	core/a380x-cockpit	master	5
20	A32NX exterior spoilers by panel group (PDF)	A32NX	core/a32nx-exterior	master	1
21	A380X electrical alerts and AC ESS loads (PDF)	A380X	core/a380x-elec	core/gps	4
22	A380X flyPad failures (PDF)	A380X	core/failures-a380x	master + core/fire + core/ecam-fws	18

23	A32NX flyPad failures (PDF)	A32NX	core/failures-a32nx	core/gps + core/fire	17
24	Display unit failures and reconfiguration (PDF)	A32NX and A380X	core/display-unit-failures	core/nd-weather-terrain	4
25	A380X weather radar GAIN, ELEVN and TILT (PDF)	A380X	core/a380x-wxr-controls	core/nd-weather-terrain + core/a380x-mfd-fms	4
26	A380X VD runways as flat ground (PDF)	A380X	core/a380x-vd-runways	core/nd-weather-terrain + core/oans-btv-raas	1
27	FCOM audit fixes (PDF)	A32NX and A380X	core/fcom-audit-fixes	master	8
28	RAT ground stow (PDF)	A32NX and A380X	core/rat-ground-stow	master	1

2. Stacks: what must be merged first

Most branches are based directly on `master` (FlyByWire master 255546cb0 merged in) and can be merged in any order. The others are stacked: a branch needs every branch under it merged first.

Stack	Merge order
A380X MFD	core/surveillance > core/a380x-mfd-fms > then any of core/a380x-atccom-datalink, core/fuel, core/gps, core/a380x-interactive-nd, core/efb-performance
Printer	... core/a380x-atccom-datalink > core/printer
flyPad	... core/efb-performance > core/efb-ground > core/efb-redesign
GPS	... core/gps > core/a32nx-air-conditioning, core/a380x-elec, core/failures-a32nx (which also merges core/fire)
A380X failures	core/fire + core/ecam-fws > core/failures-a380x (merged in 2026-10-05 for the engine failures)
ND	core/nd-weather-terrain > core/display-unit-failures; core/nd-weather-terrain + core/a380x-mfd-fms > core/a380x-wxr-controls; core/nd-weather-terrain + core/oans-btv-raas > core/a380x-vd-runways
Independent (on master)	core/a32nx-mcdu, core/ecam-fws, core/oans-btv-raas, core/isis, core/lights, core/fire, core/a380x-cockpit, core/a32nx-exterior, core/fcom-audit-fixes, core/rat-ground-stow
Feature split over two branches	The RAT stow row of core/efb-redesign needs core/rat-ground-stow (the systems side): merge both for the feature to work
Same panel.cfg section	core/display-unit-failures (A32NX DuMessage gauge) and core/efb-redesign (A32NX F/O flyPad) both add [VCockpit21]: whichever is merged second rennumbers its section to [VCockpit22]
Shared engine failure module	core/failures-a32nx and core/failures-a380x both carry the same fbw-common engine failure files (engine_failure.rs, ata.ts): whichever is merged second keeps one copy; in failures/mod.rs and the A380X a380_systems_wasm failure list keep the GPS entries of core/gps followed by one ATA 72 block

3. Companion changes in aircraft-large-files

Some features change the 3D cockpit models, which live in the separate FlyByWire repository `aircraft-large-files`. Their companion commits are on the fork `TCdr/aircraft-large-files` and must be merged with the core branch.

Companion commit	What it adds	Core branch
5563e31	A32NX MCDU screens: both MCDUs on one texture (UV halves)	core/a32nx-mcd�
36b4bd9	A380X pedestal printer paper	core/printer
27d6123	A380X printer door, roller unit and paper slit	core/printer
99237fb	A32NX printer door	core/printer
24f7b02	A380X F/O flyPad screen (independent flyPads)	core/efb-redesign
d3060ea	A32NX F/O flyPad screen (independent flyPads)	core/efb-redesign
8c14d21	A380X printer ON light (green window)	core/printer
fa6c5d1	A380X fuel jettison plume at the outlets (stacked on 8c14d21)	core/fuel
b8fb739	A380X MFP (multi-function probe) vane animations	core/a380x-cockpit
b790e8d	A32NX DU reconfiguration: 17 overlay glasses on the EIS screens, DU_MESSAGE and DU_BLANK materials (stacked on 99237fb and d3060ea)	core/display-unit-failures

4. How to read them

- Each manual has one numbered section per **feature** of its branch. The grey line under the feature title gives the aircraft, where to find it and its **status** (sim-tested, or not, with the date).
- **What it is for**: the problem the change solves, in a few sentences.
- **How to use**: numbered steps for normal use.
- **How to test**: a table of steps a tester follows in the sim, each with the expected result. If the sim does something else, it is a bug worth reporting with the branch name.
- **What you should see** tables: the expected behaviour in each situation; **Limitations and why**: what does not work yet, the reason and what to do about it.
- Pictures that do not come from the harness (drawn diagrams, in-sim screenshots taken during development) carry an amber line under their caption: they are not representative of the in-game UI and serve as a reference for the explanation only.